

Course Overview



Course Title: Database Systems (CS-207)



Instructor: Engr. Muhammad Younis



Credit Hours: 3+1



Email: myounis.faculty@aror.edu.pk

Course Guidelines

Marks Distribution

Nature of Examination (Theory)	Max. marks
Sessional (quiz, assignment, Presentation)	30
Midterm Examinations	30
Final Examinations	40
Total marks	100

Nature of Examination (Lab/Practical)	Max. marks
Manuals, Project etc	15
Midterm Examinations	15
Final Exam	20
Total marks	50

Course Objectives



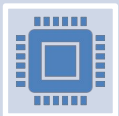
Understand fundamental database concepts.



Design database schemas using conceptual, logical, and physical models.



Apply normalization techniques to resolve database anomalies.



Use SQL for database manipulation and queries.

**In this
Lecture you
will Learn
about:**

Data

Information

Metadata

Data versus Information

Data

- Data is a plural form of *datum*, which means a piece of information.
- Data is a collection of facts & figures, such as values or measurements.
- Data can be,
 - Numbers
 - Words
 - Measurements
 - Media
- A collection of data which conveys some meaningful idea is called information.

Data (cont'd)

- Another definition of data can be a representations of objects and events that have meaning and importance in the user's environment and stored on computer media.
- For example, in a salesperson's database, the data would includes facts such as customer name, address and telephone number.
- This type of data is called *Structured Data*.
- Structured data are stored in tabular form (in tables, relations, arrays, spreadsheets, etc) and are most commonly found in traditional databases.

Data (cont'd)

- Today's databases are used to store objects such as documents, maps, photographic images, sound and video segment in addition to structured data.
- For example, the salesperson's database might include a photo image of the customer contact. It might also include a sound recording or video clip about the most recent product. This type of data is referred to as *Unstructured Data*, or as *Multimedia Data*.
 - *No fixed Schema, Varied Formats, Multiple Data Types*
- Multimedia data are most often found on Web servers and on Web-enabled databases.

Data (cont'd)



Information & Metadata

- *Information* is a data that have been processed in such a way as to increase the knowledge of the person.
- When data is collated or organized into meaningful form, it gains significance, and that meaningful data is called information.
- *Metadata* is a data about data.
- *Metadata* is data that describe the properties or characteristics of end-user data, and the context of that data. Some of the properties that are typically described include data names, definitions, size and allowable values.
- For example if *ID* is a data then the *ID* must have data type.

Information & Metadata

- **Metadata Examples:**

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- Example: Email Metadata**

- For an email:

1. **Sender:** "JohnDoe@example.com"

2. **Recipient:** "JaneSmith@example.com"

3. **Subject:** "Meeting Agenda"

4. **Date:** "January 20, 2024"

5. **Attachments:** "Agenda.pdf"

6. **Size:** "500 KB"

7. **Status:** "Unread"

8. **Importance:** "High"

- Here, the email is the primary data, and metadata includes details about the sender, recipient, subject, date, attachments, size, status, and importance.

Information & Metadata

- **Metadata Examples:**

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- Example: Student Database Metadata**

- For a student database:

1. **Table:** "Students"

2. **Columns:** "StudentID," "Name," "Age," "Grade"

3. **Primary Key:** "StudentID"

4. **Indexes:** "Name"

5. **Created:** "May 3, 2022"

6. **Updated:** "August 20, 2022"

7. **Owner:** "DBAdmin"

- In this brief example, the metadata includes the table name, column details, primary key, indexes, creation and update dates, and the owner of the database

Data Vs Information

- The terms data and information are closely related and in fact are often used interchangeably.
- However, it is useful to distinguish between data and information.
- For example
 - 221115 is a data;
 - it can be information if 22/11/15 date of your exam.
 - 221115 can be a zip code of any city.
 - PKR 2,21115 can be Salary of any Person

Structured Data vs. Information

Structured data

- Structured data refers to organized and formatted data that follows a predefined schema. It is typically stored in databases, spreadsheets, or other structured formats.
- Examples include tables in relational databases, spreadsheets with rows and columns, or data in CSV file.

Information

- Information is the result of processing and analyzing structured data to derive meaning, context, and insights.
- Information provides knowledge and understanding that can be used for decision-making or gaining insights into a particular subject.

Structured Data vs. Information

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- **Example:**
 - **Structured Data:** A database table containing customer information with columns like CustomerID, Name, Address, and PurchaseAmount.
 - **Information:** Deriving insights, such as the average purchase amount per customer, identifying trends in purchasing behavior, or segmenting customers based on specific criteria.

Tasks



1. Data vs. Info example
2. MetaData Examples

Introduction to Databases

- **What is a Database?**
 - Structured collection of data stored electronically for easy access and management.
 - A database is an organized collection of data that is stored electronically in a structured format. It allows for efficient storage, retrieval, and management of data for various purposes.
 - Think of a database as a digital filing cabinet where information (files) is stored in labeled folders for quick retrieval and efficient organization.
 - Examples: Library catalogs, online shopping platforms, social media platforms.

Introduction to Databases

1. Library

- **Analogy:** A library contains thousands of books on different subjects. Without a catalog system, finding a book would be chaotic.
- **Database Concept:** The catalog system is a database that organizes books by title, author, and subject to make retrieval easy.

2. Supermarket Inventory

- **Analogy:** A supermarket tracks the products it sells, their quantities, and prices. Without organization, they wouldn't know when to restock or update prices.
- **Database Concept:** Inventory systems are databases that store and manage product data to ensure smooth operations.

Introduction to Databases

3. Online Shopping Websites

- **Analogy:** Platforms like Amazon store data about millions of products, customers, orders, and payments.
- **Database Concept:** These platforms use databases to manage and retrieve product information, customer details, and order histories.

4. Banking System

- **Analogy:** Banks store customer details, account balances, and transaction histories. Without a database, tracking these details would be impossible.
- **Database Concept:** Banking systems use databases to manage account data securely and efficiently.

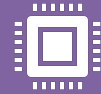
Database Management Systems (DBMS)



Definition: Software that interacts with users, applications, and the database itself to store, retrieve, and analyze data.



Functions of DBMS:



- Data storage, retrieval, and update.



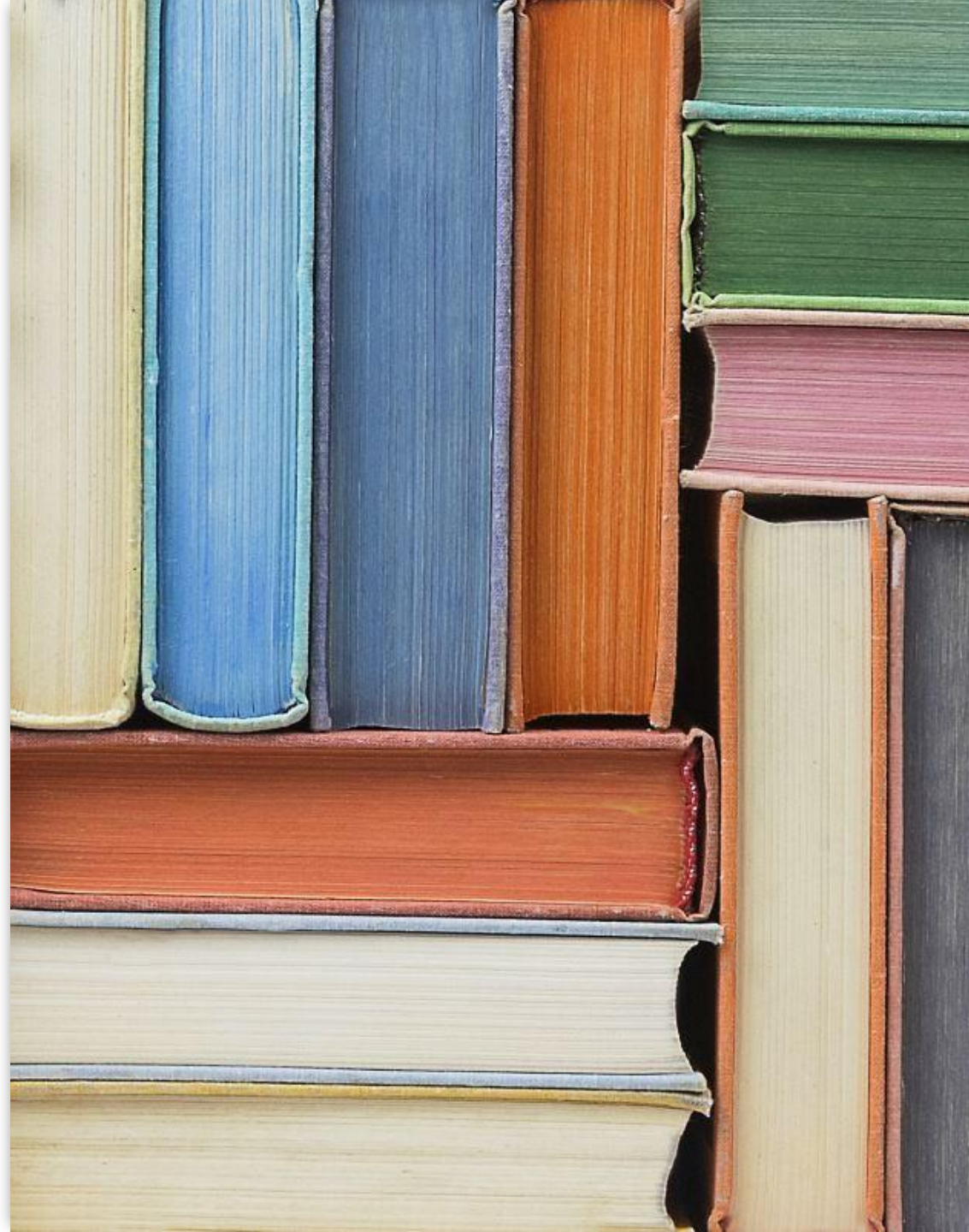
- User and security management.



- Data integrity and transaction management.

Analogy for Database Management System (DBMS):

- A **DBMS** can be compared to a **library management system** that helps organize, manage, and access books efficiently. Here's how the analogy works:
 1. **Database as a Library:** The **database** is like the collection of books in the library. It contains all the information (data) stored systematically.
 2. **DBMS as the Librarian:** The **DBMS** acts like the **librarian**, who:
 - Organizes the books on shelves (stores data in tables).
 - Keeps track of which books are where (indexes and metadata).
 - Provides a system for borrowing or returning books (data retrieval and updates).
 - Ensures books are not misplaced or damaged (data integrity).



Analogy for Database Management System (DBMS):

3. **Users as Library Visitors:** The **users** are like the people visiting the library. They can:
 - Browse for specific books (query data).
 - Borrow books or return them (insert or update data).
 - Ask for help from the librarian to find what they need (use DBMS features).

4. **Query Language as the Library Catalog:** The **query language** (e.g., SQL) is like the **library catalog**:
 - Users can search by book title, author, or subject (filter data based on conditions).
 - The librarian uses this catalog to locate books quickly.





Analogy for Database Management System (DBMS):

5. **Security and Permissions:** Just as not everyone can enter certain parts of a library (e.g., staff-only areas or rare book collections), the **DBMS** enforces access controls and permissions to ensure only authorized users can access specific data.
 6. **Backup as Book Duplicates:** In case of fire or loss, libraries often have backup copies of rare or important books. Similarly, the **DBMS** ensures data safety by maintaining backups.
 7. **Scalability as Library Expansion:** As the library grows and needs more space, the librarian organizes the collection to make room for new books. Similarly, a **DBMS** scales to handle increasing amounts of data efficiently.
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Popular DBMS



MySQL: Open-source RDBMS, commonly used for web applications.



PostgreSQL: Advanced RDBMS with strong community support.



MongoDB: A NoSQL database ideal for unstructured data.



Types of Databases: Relational Database

- Relational databases store data in tables (relations) consisting of rows and columns.
 - Relationships between tables are established using primary and foreign keys.
 - SQL is used for querying and managing data.
 - They are best suited for structured data and business applications.
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Types of Databases: Hierarchical Database

- Hierarchical databases organize data in a tree-like structure.
 - Each child record has exactly one parent record.
 - This model is simple and fast for hierarchical data but lacks flexibility.
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Types of Databases: Network Database

- Network databases extend the hierarchical model by allowing records to have multiple parents.
 - Data is represented as a graph structure.
 - This model is complex to design.
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Types of Databases: Object-Oriented Database

- Object-oriented databases store data as objects, similar to object-oriented programming.
 - They support classes, objects, inheritance, and encapsulation.
 - Used for complex data such as multimedia.
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Types of Databases: NoSQL Databases

- NoSQL databases are designed for large-scale, distributed, and unstructured data.
 - They do not strictly follow relational rules.
 - Types include key-value, document, column-family, and graph databases.
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Types of Databases: Distributed Database

- A distributed database stores data across multiple physical locations but appears as a single database.
 - It improves availability, reliability, and performance.
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Types of Databases: Cloud Database

- Cloud databases are hosted on cloud platforms and accessed via the internet.
 - They provide scalability, flexibility, and reduced maintenance.
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Applications of Databases



Banking: Account management and transactions.



Healthcare: Patient records, prescriptions.



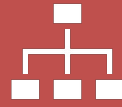
E-commerce: Product catalogs, orders.



Education: Student records, course management.

Data Models

A **data model** is a conceptual framework or blueprint that defines how data is organized, stored, and related within a database.



Hierarchical Model: Data organized in a tree-like structure.



Network Model: Data organized in a graph-like structure with nodes and edges.



Relational Model: Uses tables (relations) with rows and columns.



Object-Oriented Model: Integrates object-oriented programming and databases.

Integrates programming and databases.

Summary

Databases store and manage data for easy retrieval and manipulation.

Types of databases include relational, hierarchical, and NoSQL.

DBMS enables efficient interaction with databases.

Data models define how data is structured and accessed.